



December 2022

Duration: 3 Hours

Max. Marks:100

Note:

Part – B: Descriptive Answer type Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- The one, which converts chemical energy into electrical energy**
A) Galvanic Cell
B) Electrolytic Cell
C) Anode
D) Cathode
- Calomel electrode is an example for**
A) Metal-metal ion electrode
B) Metal-metal salt ion electrode
C) Ion selective electrode
D) Primary reference electrode
- pH of the solution can be determined using the following electrode combination.**
A) Glass-calomel electrode
B) Glass-Zinc electrode
C) Platinum-Calomel electrode
D) Zinc-Calomel electrode
- Voltage available from a battery depends upon**
A) Difference in the standard electrode potential
B) Conductivity of the electrolyte
C) Reaction quotient for the cell reaction
D) All the above
- Which of the following is a right Anode-Cathode-Electrolyte combination of a lithium-ion battery**
A) Lithium intercalated Graphite-LiCoO₂-LiClO₄ in ether
B) Lithium intercalated graphite-LiNiO₂-aqueous KOH
C) Lithium-Copper-LiBF₄ in ether
D) VH₂-LiMn₂O₄-LiPF₆ in ether
- The rusting iron is the**
A) Oxidation corrosion
B) Liquid metal corrosion
C) Wet corrosion
D) Corrosion by other gases
- When the hydrogen overvoltage on the metal surface is high, cathodic reaction is**
A) Higher
B) Slower
C) Constant
D) All of these
- type of coating is used in the preparation of printed circuit boards**
A) Electroplating
B) Electroless plating
C) Inorganic coating
D) Organic coating
- A simple molecule having two or more bonding sites through which each can link to other to form a polymer chain is called**
A) Polymer
B) Monomer
C) Initiator
D) All of these
- What is the disadvantage of bulk polymerization?**
A) Uncontrolled temperature rise
B) Difficulty in constant stirring of reaction mixture
C) Thermal degradation, discolouration of products
D) All of these
- Natural rubber is the polymerized form of**
A) Neoprene
B) Isoprene
C) Guttapercha
D) Chloroprene
- A plastic which can be softened on heating and hardened on cooling is called**
A) Thermoelastic
B) Thermoplastic
C) Thermosetting
D) Thermite

13. An example for dissolved impurity is
 A) CO_2 B) HCO_3^-
 C) NaCl D) All of these
14. Which of the following statements is correct for Winkler's method?
 A) Azides are added to destroy nitrates B) The colour change at end-point is blue to colourless
 C) Winkler's method is used to determine the amount of dissolved gases in water. D) Winkler's method is an example of iodometric titration.
15. Sludges are precipitate found boilers.
 A) Soft; adhered on the inner walls of B) Hard; adhered on the inner walls of
 C) Soft; loosely scattered in D) Hard; loosely scattered in
16. CVD method is employed to synthesize
 A) Metal oxide nanoparticles B) CO and CO_2
 C) Carbon nanotubes D) N_2
17. In presence of which gas is the fuel burnt to generate energy in form of heat?
 A) Oxygen B) Hydrogen
 C) Methane D) Nitrogen
18. Which one of the following is an example for secondary fuel?
 A) Coal B) Natural gas
 C) Kerosene D) All of these
19. In general, Liquid crystal state is also called as
 A) Triple phase B) Nematic phase
 C) Mesophase D) Polymer dispersed phase
20. Pitch is measured in case of liquid crystals
 A) Smectic B) Nematic
 C) Cholesteric D) Polymer dispersed

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1.	a) Derive the Nernst equation for single electrode potential.	6	L2	1	1
	b) Give the construction and working of the calomel electrode.	6	L2	1	1
	c) For the cell, $\text{Fe} / \text{Fe}^{+2} (0.015\text{M}) // \text{Ag}^+ (0.13\text{M}) / \text{Ag}$, Write the cell reactions and calculate the emf of the cell at STP if the standard electrode potentials of Fe and Ag are -0.44V and 0.80V respectively.	4	L3	1	2
2.	a) What is corrosion? Explain the electrochemical theory of corrosion by taking iron as an example.	8	L2	1	1
	b) Explain the two techniques of cathodic protection.	6	L2	1	1
	c) The anodic metal coating is called sacrificial coating. Justify.	2	L5	1	1
3.	a) Explain the construction and working of Li-ion battery.	6	L2	1	1
	b) What is polarization? Explain any two factors affecting polarization.	3	L2	1	1
	c) What is electroless plating? Explain the electroplating of chromium.	7	L2	1	1
Unit - II					
4.	a) What is glass transition temperature (T_g)? Explain four factors affecting T_g .	5	L2	2	1
	b) Explain the synthesis, properties, and applications of the following polymers i) polycarbonate ii) carbon fibre.	9	L2	2	1
	c) Thermosetting polymers do not exhibit plastic deformation. Justify.	2	L5	2	1
5.	a) Describe the determination of total hardness of water by EDTA method.	6	L2	2	1
	b) Explain the hot lime-soda process for softening hard water.	7	L2	2	1
	c) Explain the causes of boiler corrosion.	3	L2	2	1

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|-------|---|---|----|---|---|
| 6. a) | Explain the principle involved in the determination of dissolved oxygen by Winkler's method. | 4 | L2 | 2 | 1 |
| b) | Define the following i) graft copolymer, ii) syndiotactic polymer, iii) degree of polymerization, iv) conducting polymer. | 4 | L1 | 2 | 1 |
| c) | Describe microwave and sol-gel methods for the preparation of nanoparticles. | 8 | L2 | 2 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Explain the determination of the calorific value of solid fuel using a Bomb calorimeter. | 8 | L2 | 3 | 1 |
| b) | Explain the nematic, smectic, and columnar phases of liquid crystals. | 8 | L2 | 3 | 1 |
| 8. a) | On burning 0.83×10^{-3} kg of solid fuel in a bomb calorimeter, the temperature of 3.5 kg of water is increased from 26.5°C to 29.2°C . The water equivalent of the calorimeter and latent heat of steam is 0.385kg and $587 \times 4.2\text{kJ/kg}$ respectively. Specific heat of water is $4.2\text{kJ/kg}^{\circ}\text{C}$. The fuel contains 0.7% of hydrogen. Calculate its Gross and Net calorific values. | 5 | L3 | 3 | 2 |
| b) | Explain the mechanism of petrol knocking. | 5 | L2 | 3 | 1 |
| c) | Describe with an example, the liquid crystalline behavior of compounds based on their chemical constitutions. | 6 | L2 | 3 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
